

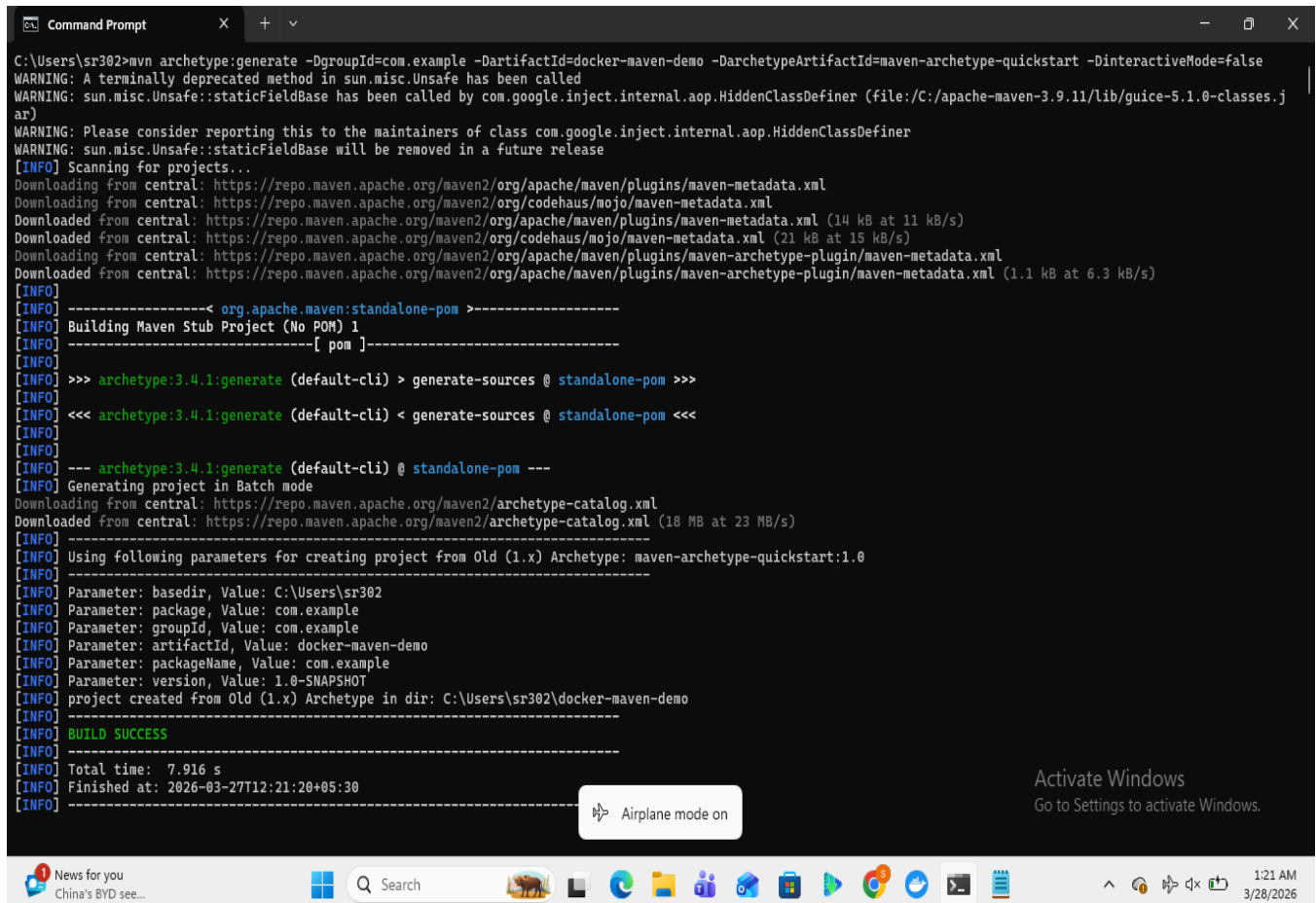
INT-332 Practical-2

Maven Practical

Exercise 1: Maven + Docker Integration.

Step 1: Create Maven Project

`mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false`



```
Command Prompt
C:\Users\sr302>mvn archetype:generate -DgroupId=com.example -DartifactId=docker-demo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 11 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metadata.xml (1.1 kB at 6.3 kB/s)
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 23 MB/s)
[INFO] -----
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: C:\Users\sr302
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: docker-maven-demo
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\sr302\docker-maven-demo
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.916 s
[INFO] Finished at: 2026-03-27T12:21:20+05:30
[INFO] -----
Activate Windows
Go to Settings to activate Windows.
Airplane mode on
```

Step 2: Package Application:-

Mvn clean package

```
C:\Users\sr302>cd docker-maven-demo

C:\Users\sr302\docker-maven-demo>mvn clean package
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/gui
ce-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:docker-maven-demo >-----
[INFO] Building docker-maven-demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ docker-maven-demo ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ docker-maven-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sr302\docker-maven-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ docker-maven-demo ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
    not setting the bootstrap class path may lead to class files that cannot run on JDK 8
    --release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:options.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ docker-maven-demo ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
```

```

[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ docker-maven-demo ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
not setting the bootstrap class path may lead to class files that cannot run on JDK 8
--release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ docker-maven-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit.JUnit3Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.120 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ docker-maven-demo ---
[INFO] Building jar: C:\Users\sr302\docker-maven-demo\target\docker-maven-demo-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 5.692 s
[INFO] Finished at: 2026-03-27T12:21:47+05:30
[INFO]
[INFO] -----

```

Step 3: Modify Java Program:-

With path : src/main/java/com/example/App.java



```
pom.xml App.java Dockerfile
File Edit View

package com.example;

public class App
{
    public static void main(String[] args)
    {
        System.out.println("Hello from Maven + Docker Integration!");
    }
}
```

Step 4: Create Dockerfile:-



```
pom.xml App.java Dockerfile
File Edit View H1 ⋮ B I S ↺

FROM eclipse-temurin:21-jdk

WORKDIR /app

COPY target/docker-maven-demo-1.0-SNAPSHOT.jar app.jar

ENTRYPOINT ["java", "-jar", "app.jar"]
```

Step 4: Add dockerfile-maven-plugin in pom.xml:-

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<groupId>com.example</groupId>
```

```
<artifactId>docker-maven-demo</artifactId>
```

```
<version>1.0-SNAPSHOT</version>
```

```
<packaging>jar</packaging>
```

```
<name>docker-maven-demo</name>
```

```
<properties>
```

```
<project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
```

```
<maven.compiler.source>21</maven.compiler.source>
```

```
<maven.compiler.target>21</maven.compiler.target>
```

```
</properties>
```

```
<dependencies>
```

```
<dependency>
```

```
<groupId>junit</groupId>
```

```
<artifactId>junit</artifactId>
```

```
<version>4.13.2</version>
```

```
<scope>test</scope>
```

```
</dependency>
```

```
</dependencies>
```

<build>

<plugins>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-jar-plugin</artifactId>

<version>3.3.0</version>

<configuration>

<archive>

<manifest>

<mainClass>com.example.App</mainClass>

</manifest>

</archive>

</configuration>

</plugin>

<plugin>

<groupId>com.spotify</groupId>

<artifactId>dockerfile-maven-plugin</artifactId>

<version>1.4.13</version>

<configuration>

<repository>docker-maven-demo</repository>

<tag>\${project.version}</tag>

</configuration>

<executions>

<execution>

<id>default</id>

<goals>

<goal>build</goal>

</goals>

</execution>

</executions>

</plugin>

</plugins>

</build>

</project>

Step 5: Build Docker Image Using Maven:-

mvn dockerfile:build

```
Command Prompt
C:\Users\sr302\docker-maven-demo>mvn dockerfile:build
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/apache-maven-3.9.11/lib/gui
ce-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO] -----< com.example:docker-maven-demo >-----
[INFO] Building docker-maven-demo 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- dockerfile:1.4.13:build (default-cli) @ docker-maven-demo ---
[WARNING] 1 problem was encountered while building the effective model for org.javassist:javassist:jar:3.18.1-GA during dependency collection step f
or plugin (use -X to see details)
[INFO] dockerfile: null
[INFO] contextDirectory: C:\Users\sr302\docker-maven-demo
[INFO] Building Docker context C:\Users\sr302\docker-maven-demo
[INFO] Path(dockerfile): null
[INFO] Path(contextDirectory): C:\Users\sr302\docker-maven-demo
[INFO] Image will be built as docker-maven-demo:1.0-SNAPSHOT
[INFO] Step 1/4 : FROM eclipse-temurin:21-jdk
[INFO] Pulling from library/eclipse-temurin
[INFO] Digest: sha256:3b0a98dfbdf1067c20a7854cec159551777d2ee1381bc76cd4bd0719f543b148
[INFO] Status: Image is up to date for eclipse-temurin:21-jdk
[INFO] ----> 3b0a98dfbdf1
[INFO] Step 2/4 : WORKDIR /app
[INFO] ----> Using cache
[INFO] ----> 8e1da7d27c53
[INFO] Step 3/4 : COPY target/docker-maven-demo-1.0-SNAPSHOT.jar app.jar
[INFO] ----> Using cache
[INFO] ----> c786a85ee644
```

```

[INFO] ---> Using cache
[INFO] ---> 8e1da7d27c53
[INFO] Step 3/4 : COPY target/docker-maven-demo-1.0-SNAPSHOT.jar app.jar
[INFO] ---> Using cache
[INFO] ---> c786a85ee644
[INFO] Step 4/4 : ENTRYPOINT ["java","-jar","app.jar"]
[INFO] ---> Using cache
[INFO] ---> 78f359f67482
[INFO] Successfully built 78f359f67482
[INFO] Successfully tagged docker-maven-demo:1.0-SNAPSHOT
[INFO] Detected build of image with id 78f359f67482
[INFO] Successfully built docker-maven-demo:1.0-SNAPSHOT
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 10.766 s
[INFO] Finished at: 2026-03-27T12:36:07+05:30
[INFO] -----

```

Step 6: Verify Docker Image:-

Docker images

```

C:\Users\sr302\docker-maven-demo>docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-maven-demo:1.0-SNAPSHOT	78f359f67482	675MB	211MB	
eclipse-temurin:21-jdk	3b0a98dfbdf1	679MB	214MB	
flask-user-app:latest	d518b01bf622	210MB	51.1MB	
ghcr.io/rajjadav23/flask-ghcr:v1	eedb6de58305	203MB	49.7MB	
ghcr.io/rajjadav23/ghcr-demo:latest	6536900dd62a	92.5MB	26MB	
ghcr.io/rajjadav23/mysql-college:v1	6b4a7862ee90	1.07GB	232MB	
hello-world:latest	05813aedc15f	25.9kB	9.52kB	
httpd:latest	b89c19a39051	177MB	47.6MB	U
mongo:6	03cda579c8ca	1.06GB	273MB	U
mongo:latest	474f5c3bf0e3	1.29GB	338MB	
mysql:8.0	a3dfff78d8762	1.08GB	247MB	U
nginx:latest	c881927c4077	240MB	65.7MB	U
node-mongo-compose-app:latest	132c7889795c	1.63GB	406MB	U
php_4705:latest	5f141c668873	707MB	176MB	U
python-flask-app:latest	a11e5a4931bf	210MB	51.1MB	
raaz01/php_4705:v1	4be8740fbe76	707MB	176MB	U
raaz01/ubuntu:v1.1	a4453623f2f8	117MB	29.7MB	U
redis:latest	315270d16608	204MB	55.3MB	U
ubuntu:latest	cd1dba651b30	119MB	31.7MB	U
wordpress:latest	ba1996f128e9	1.08GB	275MB	U

Step 7: Run Docker Container:-

docker run docker-maven-demo:1.0-SNAPSHOT

```

C:\Users\sr302\docker-maven-demo>docker run docker-maven-demo:1.0-SNAPSHOT
Hello from Maven + Docker Integration!

```